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A Preschooler With Severe Social Anxiety and Selective Mutism Versus Autism Spectrum Disorder

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ABSTRACT

Selective mutism (SM) and severe social anxiety can mimic autism spectrum disorder (ASD) when assessment relies heavily on observations in unfamiliar contexts. Multidisciplinary assessments comparing social interaction and communication across familiar and unfamiliar contexts may help prevent misdiagnosis. While autism-like behaviors are restricted to anxiety-provoking contexts in SM and severe social anxiety, ASD is characterized by pervasive social deficits across relationships and settings.

1 | Introduction

Selective mutism (SM) is a childhood psychiatric condition defined by a functionally impairing failure to speak in specific social situations despite the ability to speak in other contexts. The condition must be at least one month in duration, not limited to the first month of school, not attributable to lack of comfort with the required language, and not occurring exclusively during the course of a psychotic disorder or autism spectrum disorder (ASD). The prevalence of SM is thought to be 0.7%–2.0%, varying depending on samples and diagnostic methods [1]. Its onset is around 2.7–4.5 years of age [2, 3], but it is often not detected until school age, with some studies documenting diagnosis as late as 8.8 years [3]. Studies of outcomes in children with SM suggest that it resolves over time. One review found that 190 of 243 patients with SM showed moderate or total improvement at follow-up intervals ranging from 2 to 17 years, with other anxiety disorders being the most common psychiatric diagnoses at the follow-up [4].

Children with SM may present similarly to those with ASD in certain social situations, as they may not speak, may interact less, become anxious, appear shy or uncomfortable, show reduced eye contact and facial expressions, and become withdrawn in group activities. There currently is no consensus or guideline on how to differentiate SM vs. ASD. Many clinicians lack familiarity with the topic, which leads to misdiagnosis and delays in targeted therapy. Here, we present the case of a preschool-aged child evaluated at the autism specialty center. He was evaluated by the multidisciplinary team and ultimately was diagnosed with SM.

2 | Case Presentation/Examination

The patient was a 5-year-old boy referred to the autism specialty center due to his parents' concern for anxiety and shyness that in turn impacted his social communication skills and led to difficulty with unexpected changes. Parents initially became concerned at age 2–3 as he was not speaking or interacting at day care and was shy with strangers. There were no concerns about motor or speech/language development.

Medical history was notable only for seasonal allergies. The patient was attending kindergarten and lived with immediate family. Family history was notable for psychiatric and

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TABLE 1 | Selective mutism questionnaire and preschool anxiety scale.

Selective mutism questionnaire	School at the beginning	School, currently (0–18)	Home/family (0–15)	Public/Social (0–15)
Score	5*	12	8	3*
Score	4*	14	11	4*
Preschool anxiety scale-parent		**Parent**		**Parent**
Total *T* score		67*		86**
OCD *T* score		57		67*
Social anxiety *T* score		71**		76**
Separation anxiety *T* score		46		68*
Physical injury fears *T* score		68*		87**
General anxiety *T* score		66*		86**

Note: Lower score: lower frequency of speaking behavior, more severe symptoms.
* and **Indicate significant findings.

learning-related conditions. Birth history was significant for neonatal jaundice treated with phototherapy.

The patient had exhibited social anxiety and stranger fears since 2–3 years of age. In the presence of strangers, unfamiliar people or places, public places, and loud environments, he would become restless and pace, become very quiet, or have tantrums. At the beginning of the school year, he was shy, anxious, and quiet. It took him months to speak at school. He avoided interaction with others, stayed by himself in class, and stayed behind the curtain. At the time of presentation, he and his parents had been attending the same religious group for 3 years. During the first 6–8 months, he did not speak at all. He continued not to engage in group activities. He typically would not speak at gatherings of relatives. He had several "comfortable" people with whom he interacted, including mom, dad, sibling, maternal grandmother, paternal grandparents, maternal aunt, current teacher, and 3–4 school peers. With them, he would speak well and make good turn-taking conversations. He had psychotherapy with components of play therapy and family therapy for about 6 months prior to presentation, and he would speak in sessions. He had fears of heights, crowded places, swimming, and insects. He had once seen a fallen tree and since then had fears of storms, loud noises, thunder, and rain. Each parent completed the Selective Mutism Questionnaire and the Preschool Anxiety Scale [5, 6] (Table 1).

He exhibited some restrictive and repetitive behaviors, including difficulty with changes in routines, wanting plates and cups to be in specific places at mealtimes, strong interest in video games, sensitivity to bright light and certain sounds and smells, and a behavior of rubbing his belly button. His parent completed the Sensory Processing Measure –2 Child Form [7] (Table 2). He had hearing sensitivity in the moderate-difficulty range. He was sensitive to loud sounds, crowded/noisy environments, and he startled easily to sounds. Additionally, he tended to seek movement and proprioceptive sensory feedback. He pursued activities such as pushing, pulling, dragging, lifting, and jumping. He applied too much pressure and force at times, walked loudly as if his feet were heavy, and often broke objects accidentally. He had longstanding problems with inattention, hyperactivity, and impulsivity. He was scored on the ADHD Rating Scale-IV Preschool by parents and teachers [8] (Table 3).

TABLE 2 | Sensory processing measure—2 (SPM—2) child form (5–12 years).

Domain	*T*-scores	Descriptive category
Vision	54	Typical
Hearing	61*	Moderate Difficulties
Touch	49	Typical
Taste and smell	45	Typical
Body awareness	64*	Moderate Difficulties
Balance and motion	59	Typical
Sensory total	57	Typical
Planning and Ideas	61*	Moderate Difficulties
Social participation	65*	Moderate Difficulties

Note: The sensory processing measure *T*-scores Typical Range from 40 to 59; Moderate Range 60–69; Severe Difficulties 70–80.
*Indicates significant findings.

He had tantrums, which would happen almost daily, multiple times per day, at home and in public spaces, and last 20 min to 2 h. Tantrums would consist of crying, screaming, yelling, stomping, throwing or breaking things, hitting, kicking, biting, and scratching others, hitting himself, and throwing himself on the floor. Triggers included getting anxious, not getting his way, hearing no, transitioning, and changes of routines.

The patient had no history of speech therapy, and parents reported that speech and language milestones were met on time. Parents reported strengths in the use of complex language and advanced vocabulary.

The patient had no history of occupational therapy. Parents reported strength and interest areas that included the following:

TABLE 3 | ADHD rating scale-IV preschool (ADHDRS-4P).

ADHDRS-4P	Inattention (IN) score, symptom count	Hyperactivity/ Impulsivity (HY/IM) score, symptom count	Total score	93rd % score
Parent	16 (6/9)**	21 (7/9)**	37	IN 14, HY 17, Total 32
Parent	10 (2/9)	18 (6/9)**	28	IN 14, HY 17, Total 32
Teacher	24 (9/9)**	26 (9/9)**	50	IN 18, HY 22, Total 38
Teacher	18 (8/9)**	4 (1/9)	22	IN 18, HY 22, Total 38
Teacher	11 (2/9)	12 (3/9)*	23	IN 18, HY 22, Total 38
Teacher	7 (0/9)	0 (0/9)	7	IN 18, HY 22, Total 38
Teacher	3 (0/9)	2 (0/9)	5	IN 18, HY 22, Total 38
Teacher	0 (0/9)	2 (0/9)	2	IN 18, HY 22, Total 38

* and **Indicate significant findings.

video games, playing outside, helping his father with yard work, and playing rugby. He had age-appropriate dressing skills, bathing skills, tolerated nail care, and dental hygiene and haircare were within age level expectations. Parents noted that visits to the dentist and barber were initially challenging secondary to anxiety and concerns for sound sensitivities, but accommodation (e.g., haircuts performed at home by a parent and establishing predictable dental care routines) enabled him to appropriately adapt to these hygiene tasks. He was able to self-feed and ate a variety of food tastes and textures with occasional sensitivity to certain foods. Parents did not have any concern with chewing and swallowing foods. Additionally, parents reported a well-established sleep routine throughout his life. Parents did not report any concerns about motor performance. He could ride a bike independently. He played rugby and soccer. The patient had complex symbolic play, creating stories with action figures lasting 5–8 min.

His social milestones (Table 4) have been met on time in the context of interaction with familiar people. With familiar people, at the time of presentation, he exhibited a social smile and was well engaged in turn-taking conversation. He was orienting to his name, following a point, giving objects to initiate interaction/get help. He used waving bye-bye and a variety of gestures. He showed objects to his parents to share interest. However, he did not demonstrate these social communication skills with strangers. With strangers, he did not exhibit a social smile, engage in conversation, seek play and attention, or show objects to initiate interaction of shared interest. He avoided interaction with strangers.

3 | Methods

During the speech-language evaluation, the patient participated in a standardized language assessment and the Autism Diagnostic Observation Schedule-2 (ADOS-2) Module 3. During the parental interview, he was shy and had limited interaction with the clinician. Once his parents left the room, he became more comfortable with the clinician as the session progressed and was observed to engage in back-and-forth interactions and follow directions.

Speech and language skills were assessed through clinical observation and Clinical Evaluation of Language Fundamentals-Fifth Edition [10] (Table 5). He presented with above average core language skills compared to same-age peers as he communicated using full, complex sentences for a variety of communicative functions with minimal grammatical errors. His parents also completed the Pragmatics Profile parent observation checklist, which revealed slightly below-average pragmatic language skills.

The patient participated in ADOS-2, a semi-structured assessment used to observe language/communication, reciprocal social interaction, play, imagination and other behaviors associated with ASD [9, 11] (Table 6). While it is regarded as a reference standard in ASD assessment, literature indicates that substantial false positives remain challenging for clinical practice [9, 12–14]. Based on observations during ADOS-2, he presented with moderate level of autism spectrum related symptoms. He presented with differences in social affect, including somewhat reduced shared enjoyment likely secondary to shyness, reduced use of social overtures, and reduced frequency of response to conversation bids initially. Responsiveness to social contexts was varied, though reduced, and quality of rapport with the clinician was reduced. He presented with some restricted and repetitive behaviors, including chewing on his shirt and putting his finger in his mouth, making a few references to video games, and possible hand/finger mannerisms. He also demonstrated differences in play (i.e., did not narrate play out loud, resulting in clinician missing context surrounding the play routine). Features of anxiety were observed, and he was fidgety and active in his chair throughout. He demonstrated the following characteristics not consistent with ASD: no observed echolalia, stereotyped language, self-injurious behavior or compulsions/rituals; appropriate intonation, volume and rate of speech; offering information about himself spontaneously throughout for social purposes; using a variety of gestures throughout paired

TABLE 4 | Social milestone of patient compared with typical development.

Social milestones in typical development [9]	Around 12 months old
Social smiles (8 weeks. By 3 months)	Yes
Orients to name well (9 months. By 12 months)	Yes
Follows a point, "Oh look at..." (9 months)	Yes
Engage in back-and -forth, two-way communication using vocalization and eye contact (By 9 months)	Yes
Play Peek-A-Boo (10 months. By 12 months)	Yes
Waves bye-bye (10 months. By 12 months-imitation. By 15 months-use)	Imitation
Shows objects to parent to share interest (By 12 months)	Yes
Give to initiate interaction or get help (9 months, By 12 months)	Yes

Social milestones in typical development [9]	With familiar people	With unfamiliar people
Social reciprocity and relationship		
Reciprocal smiling (2 months, by 3 months)	Yes	He avoids interaction
To & fro alternating vocalizations (4 months) Engage in back-and -forth, two-way communication using vocalization and eye contact (By 9 months)	He makes good turn-taking conversations	
Enjoy extended play with others; especially care-givers (By 9 months)	Yes	
Seeks and enjoys attention from others, especially caregivers (By 15 months)	Yes	
Engages in parallel play with peers (By 15 months)		Yes, to certain extent
Uses words to describe what someone else is thinking. "Mom thought I was asleep" (3 years)	Yes	He avoids interaction
Labels happiness, sadness, fear, & anger in self. Uses "feeling" words (4 years)	Yes	
Deception: interested in "tricking" others & concerned about being tricked by others (4 years)	Yes	
Modulates or modifies voice correctly depending on situation or listener (e.g., outside voice, to adult, other infant/young child, or younger child) (By 60 months)	Yes	
Follows a point, "Oh look at..." (By 9 months)	Yes	
Orients to name well (9 months. By 12 months)	Yes	
Give to initiate interaction or get help (9 months, By 12 months)	Yes	
Shows objects to parent to share interest (By 12 months)	Yes	
Non-verbal communication		
Eye-contact	Good	Poor
Proto-imperative pointing to get desired objects (By 12 months)	Yes	He avoids interaction
Proto-declarative pointing to express/share interest (By 15 months)	Yes	
Uses complex communication skills integrating gestures, vocalizations, and eye contact (e.g., looking to parent while taking his or her hand to bring him or her to a desired toy, By 15 months)	Yes	
Shakes head for no (8 months. By 15 months)	Yes	
Nodding head for yes (15 months)	Yes	
Waves bye-bye (10 months. By 12 months-imitation. By 15 months-use)	Yes	
Blow a kiss (13 months)	Yes	
Shh gesture (14 months)	Yes	
Thumbs up (15 months)	Yes	
High 5 (16 months)	Yes	
Other		
Stranger anxiety (At 6 months–1 year, when exposed to a stranger)		High

TABLE 5 | Clinical evaluation of language fundamentals—fifth edition (CELF-5).

Subtest	Scaled score	Percentile rank
Sentence comprehension	15	95
Word structure	13	84
Formulated sentences	15	95
Recalling sentences	17	99
Pragmatic profile	7	16

Index score	Sum of scaled scores	Standard score	Percentile rank
Core language (CLS)	60	134	99

Note: The *CELF-5* evaluates a child's performance on language measures of word meanings and vocabulary, word and sentence structure, rules of oral language used in responding to and conveying messages, and recall and retrieval of spoken language. A Core Language Score is obtained which is a measure of general language ability and is used to determine if a language disorder exists. The mean standard score is 100 with a standard deviation of 15.

TABLE 6 | Autism Diagnostic Observation Schedule-2 (ADOS-2) Module 3.

	Code 0 (no evidence of abnormalities)	Code 1 (mild abnormal or slightly unusual, not grossly abnormal)	Code 2 or 3 (definitely or markedly abnormal)
Social affect	A5. Offer information A7. Reporting of events[a] B7. Quality of social overture	A6. Ask for information A8. Conversation[a] B4. Shared enjoyment in interaction[a] B9. Quality of social response[a] B10. Amount of reciprocal social communication[a]	B8. Amount of social overture/ maintenance of attention
	A9. Gestures[a] B1. Unusual eye contact[a] B3. Language linked with non-verbal communication	B2. Facial expressions directed to examiner[a]	
	B5. Comments on others' emotion and empathy B6. Insight into typical social situations and relationship	B11. Overall quality of rapport[a]	
Restricted and repetitive behaviors	A2. Speech abnormalities with autism A3. Immediate echolalia A4. Stereotyped/idiosyncratic speech[a] D3. Self-injurious behavior D5. Compulsions or rituals	D2. Hand finer, other complex mannerisms[a] D4. Excessive interest in or references to unusual or highly specific topics or objects or repetitive behaviors[a]	D1. Unusual sensory interest*
Play		C1. Imagination/creativity	
Other behaviors	E2. Tantrums, aggression, negative, disruptive behavior	E3. Anxiety	E1. Overactivity/Agitation

ADOS-2 Module 3	Algorithm score
1. Social Affect (SA)	
(1) Communication	
A7. Reporting events	0
A8. Conversation	1
A9. Descriptive, conventional, instrumental, or informational gestures	0
(2) Reciprocal social interaction	
B1. Unusual eye contact	0
B2. Facial expressions directed to examiner	1
B4. Shared enjoyment in interaction	1

(Continues)

TABLE 6 | (Continued)

ADOS-2 Module 3	Algorithm score
B7. Quality of social overtures	0
B9. Quality of social response	1
B10. Amount of reciprocal social communication	1
B11. Overall quality of rapport	1
2. Restricted and repetitive behaviors (RRB)	
A4. Stereotyped/idiosyncratic use of words or phrases	0
D1. Unusual sensory interest in paly material/person	2
D2. Hand and finger and other complex mannerism	1
D4. Excessive interest in or references to unusual or highly specific topics or objects or repetitive behaviors	1
3. Summary	
Social affect total (SA) score	6
Restricted and repetitive behavior total (RRB) score	4
Overall total (SA+ RRB) score Autism classification cutoff: 9 Autism spectrum classification cutoff: 7	10
ADOS-2 classification	Autism classification
ADOS-2 comparison score	6
ADOS-2 level of Autism spectrum-related symptoms	Moderate

[a]Items are included in the algorithm scoring.

TABLE 7 | Beery-Buktenica developmental test of visual motor integration.

	Standard score	Percentile	Scale score	Performance
Visual motor integration	96	39	9	Average (90–109)
Visual perceptual	123	94	15	High (120–129)
Motor coordination	90	25	8	Average (90–109)

Note: This test contains three parts: Visual Motor Integration (VMI), Visual Perception, and Motor Coordination. The performance is ranked as very high, high, average, low, or very low based on the percentage of an age group that scores within a range of standard scores. The mean standard score is 100 with a standard deviation of 15.

with verbal communication and consistent eye contact as he got more comfortable with the clinician; directing a variety of facial expressions toward the clinician; spontaneously commenting on the emotions of others; and demonstrating understanding of social relationships and his role. While total score was in Autism classification with the moderate level of autism spectrum-related symptoms, many of the features that received a code of 1 were likely secondary to his shyness/reserved nature [12, 13].

During the occupational therapy evaluation, he easily followed directions. He eventually engaged briefly with the clinician, but he was very shy and initially hid behind his mother. He did not speak during the assessment; however, he followed directions and transitioned easily between tasks. Later, once testing was completed, he demonstrated an increased level of comfort with the clinician and he communicated in sentences and allowed the clinician to join his play. The Beery-Buktenica Developmental Test of Visual Motor Integration was completed [15] (Table 7). Fine motor and visual motor skills were average for his age; visual perceptual performance scored in the high range. Based on assessment scores and clinical observations, fine and visual motor skills were determined to be appropriate for his age with visual perceptual skills being above age level. No overt sensory differences were observed although sensory accommodations were embedded into the testing environment.

The patient was seen by the psychiatrist twice in two separate sessions. At the first session, he hesitated to approach the office and appeared anxious when he entered. He turned away and avoided eye contact when the psychiatrist greeted him. Behaviors included fidgeting in his chair, biting mom's bag, and hitting dad with a ball. He exhibited moderately increased motor activity. He did not engage in the interview with the psychiatrist and instead played silently with action figures. He was observed to make a story play with action figures and dollhouse even though he did not speak. He responded to joint attention. He made no speech until 25 min into the visit, at which point he

spoke a few words to his mother for a request, using an informative gesture (fingers to indicate numbers) with minimal speech. He continued to appear anxious but made eye contact with parents and spoke a few short phrases. When he wanted a snack, he whispered in mom's ear. He managed to look at the psychiatrist and nod when offered a snack. His mother prompted him to speak, and he pointed to the psychiatrist and said, "Can I have a snack?" This was his only engagement during the first session. At the second session, the patient and parents were evaluated using the Parent–Child Early Relationship Assessment (PCERA) through a 1-way mirror [16] (Figure 1). This consisted of four parts: snack time, block design task, free play, and separation-reunion procedure (Ending and Post-session were also described below because they were informative, although they are not formal parts of the PCERA).

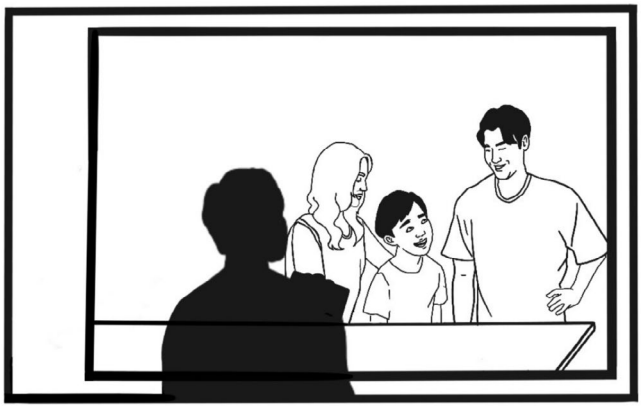


FIGURE 1 | Observation through one-way mirror.

1. Snack time: The psychiatrist entered the office and he hid himself under the desk. When the psychiatrist announced "It's snack time" and left the office, he gave a thumbs up (conventional gesture). He retrieved snack, showed it to dad, used a finger to count (informational gesture), and shared snack with parents. He continued to use conventional gestures with parents—including shrugging shoulders and nodding head—while making good eye contact and turn-taking conversation with parents. He appeared playful and comfortable with parents and was speaking throughout the time when the psychiatrist was away (Figure 2).
2. Block design: The psychiatrist reentered the office for the block design task. He ignored the psychiatrist and did not make eye contact. When the psychiatrist stepped out, however, he made eye contact with parents, nodded his head, used proximal pointing, and spoke in complex sentences. Parents provided guidance and praise while he did the block design task.
3. Free play: The psychiatrist returned to provide free-play materials. He did not acknowledge the psychiatrist when asked how he was doing. He continued putting blocks together until the psychiatrist asked to remove the blocks, at which point he helped clean up without making eye



FIGURE 2 | Snack time.



FIGURE 3 | Free play.

contact. Once the psychiatrist stepped out, he engaged in complex story play while maintaining conversation and nonverbal communication with parents. He responded immediately when mom showed him action figures (Figure 3).

4. Separation-reunion: The psychiatrist stepped in and his parents left the office telling him that they would meet another doctor. He appeared anxious with freezing/tense facial expressions and eyes wide open. When the psychiatrist attempted to engage him in play, he ignored him and

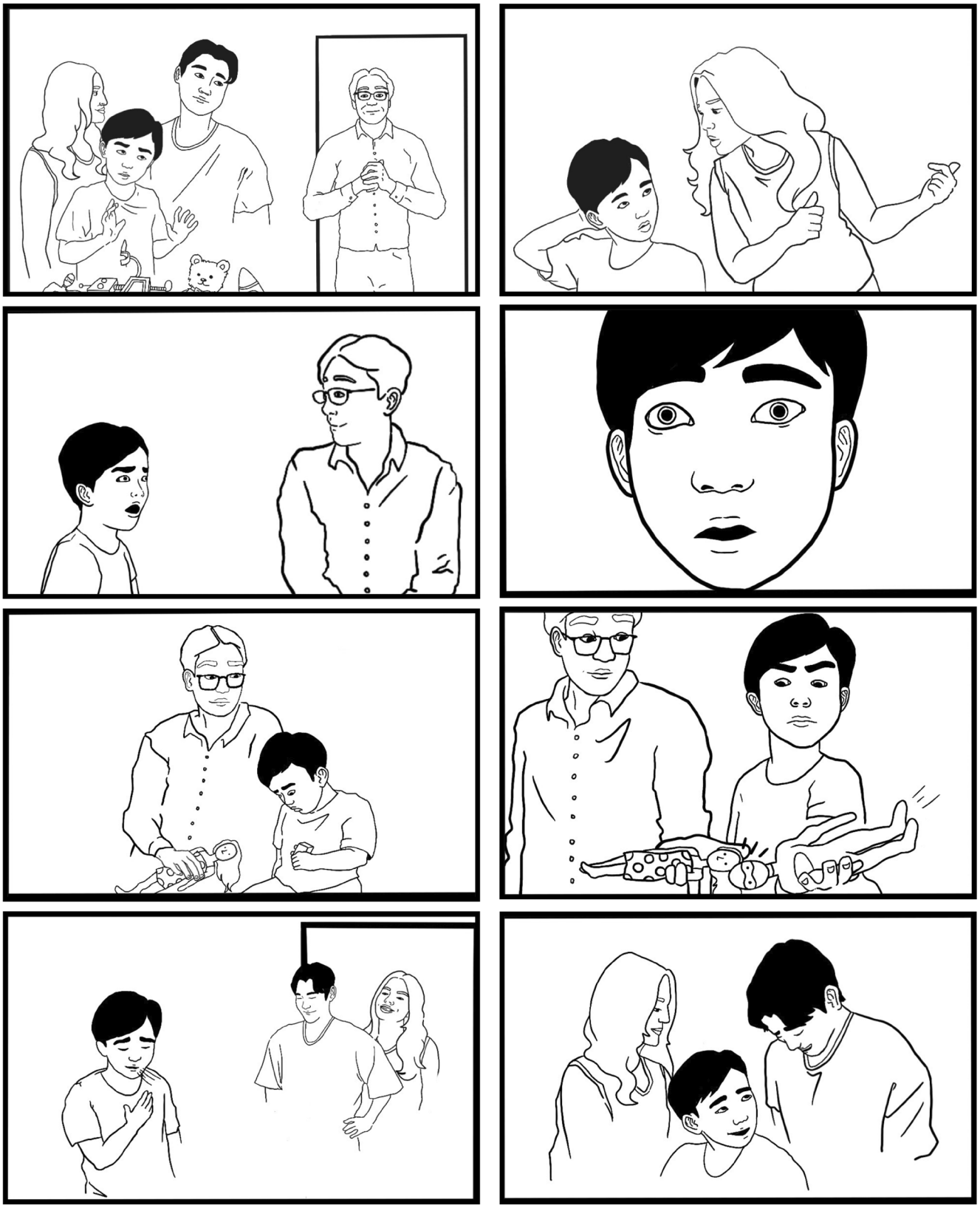


FIGURE 4 | Separation-reunion.

turned away. When the psychiatrist brought action figures to interact with his action figures, he hit and pushed away the psychiatrist's action figures. He remained mute the entire time the parents were away. The parents then returned and the psychiatrist exited. It took some time for him to begin speaking to his parents again, but he ultimately returned to smiling and making eye contact and conversation with them (Figure 4).

Ending: The psychiatrist prompted him that it was time to go home and offered a hand for a handshake. He grabbed dad's hand and placed it in the psychiatrist's hand (Figure 5).

Post-session: On the way home, parents asked how he was doing when they were out. He expressed experiencing gastrointestinal distress (Figure 6).

There was a significant discrepancy in this patient's social communication and interaction with parents only vs. parents and a stranger, as well as with parents vs. a stranger (Table 8). During the separation-reunion procedure, he experienced an acute severe anxiety attack, caused by separation from parents and staying with a stranger, manifested by eyes wide open, freezing/tense facial expressions, avoidance of interaction, being mutistic, and physical symptoms of abdominal pain and nausea.

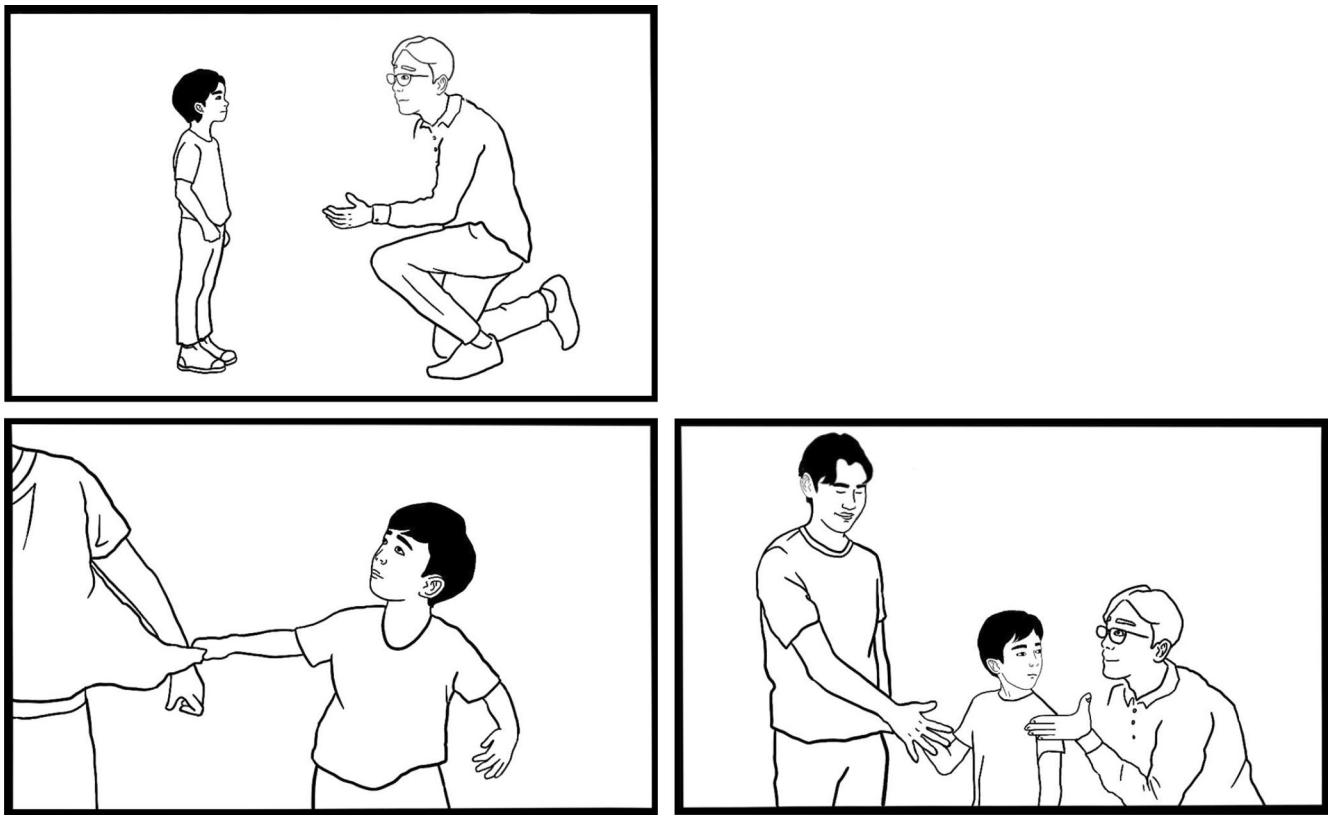


FIGURE 5 | Ending.



FIGURE 6 | Post-session.

TABLE 8 | Patient's anxiety, language, and behaviors by context.

Context Behaviors	Parents	Speech language pathologist (female, the same race)	Occupational therapist (female, different race)	Psychiatrist (male, different race)
Anxiety	No anxiety	Some anxiety (ADOS-2 anxiety code 1)	Very anxious	Extremely anxious
Language	Complex sentences	Complex sentences (ADOS-2 Module 3)	Minimally speaking	Not speaking at all
Social behaviors	Intact social interaction	Mixed features Strength in non-verbal communication	Following directions for testing	No interaction Severe stranger anxiety and distress

DSM-5 Criterion A of ASD requires persistent deficits in social communication and interaction across multiple contexts [17]. To be diagnosed with ASD, a child must demonstrate persistent deficits in social–emotional reciprocity, non-verbal communication, and developing, maintaining, and understanding social relationships across various contexts as a long-standing pattern since early childhood, including both familiar people and unfamiliar people and both home and school/community. Because his social behaviors are markedly context-dependent, with a clear contrast between familiar people and unfamiliar people (Table 8), his social difficulties are not well explained by ASD and are better explained by SM and severe social anxiety.

Final diagnosis for this patient included the following: social anxiety disorder, severe; selective mutism; specific phobia, natural environment/situation type; and other specified disruptive, impulse-control, and conduct disorder. He also received a provisional diagnosis of attention deficit hyperactivity disorder.

4 | Discussion

SM is classified as an anxiety disorder, and there is notable overlap between SM and social anxiety disorder (SAD). Some clinicians consider them to be synonymous, with SM as either an extreme version or early manifestation of SAD [18]. In one study, clinical interviews found that 69% of children with SM also had SAD [19]. Another study looked at 130 children with SM and used latent class analysis to derive three clinical subgroups based on parent reports of social anxiety, behavioral problems, and communication delay. All three groups were characterized by significant social anxiety. One group was characterized by only anxiety (12.3%), but the two larger groups were characterized by anxiety and behavioral problems (44.6%) and anxiety and language problems (43.1%) [20]. This suggests that the disorder shares features not only with anxiety disorders but also with behavioral and neurodevelopmental disorders.

The overlap between SM and ASD is complicated by the fact that diagnostic criteria preclude the diagnosis of SM "during the course of ASD," though ASD is not an episodic disorder. ASD is defined by persistent deficits in social communication and interaction along with restricted, repetitive behaviors. Studies allowing for concomitant diagnosis of ASD and SM have found that the disorders share a relationship similar to that of SM and SAD. A retrospective study found that among 97 children with SM, 67% met criteria for ASD, and another 19.6% had subclinical autistic features [3]. In children with co-occurring SM and ASD, sensory issues may mediate the relationship between the diagnosis and social anxiety. A study compared sensory behaviors in children with SM + ASD and those with SM only. They found that sensory avoidance predicted social anxiety for children in the SM + ASD group, suggesting that therapy should be adapted to target sensory processing when SM and ASD are co-diagnosed [21].

Our patient's evaluation highlights the complex relationships between ASD, SM, and SAD. Although our patient scored in the moderate level of autism spectrum-related symptoms on ADOS-2, many of these features were observed only during interactions with an unfamiliar person (an examiner). Because he demonstrated adequate social–emotional reciprocity, nonverbal communication, and social relationships in the interaction with familiar people, this elevated level of autism spectrum-related symptoms was thought to be driven by his social anxiety. Lieb et al. investigated 3364 children (2034 with ASD and 1330 without ASD) who had ADOS-2 and found that the presence of anxiety was associated with increased Social Affect severity only among children without ASD [22]. SM and social anxiety best explain this patient's constellation of symptoms. In children presenting with features of ASD, it may be important to evaluate anxiety and SM.

Author Contributions

Anna Dowling: writing – original draft. **Erin Moxey:** conceptualization, methodology, writing – original draft, writing – review and editing. **Meghan Keenan:** conceptualization, methodology, writing – original draft, writing – review and editing. **Ji Su Hong:** conceptualization, methodology, project administration, resources, supervision, writing – original draft, writing – review and editing.

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Consent

Written informed consent was obtained from the patient's legal guardian for the publication of this case report. We are grateful to the patient's legal guardian for the support in writing this manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data supporting the findings of this case report are available within this paper.

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